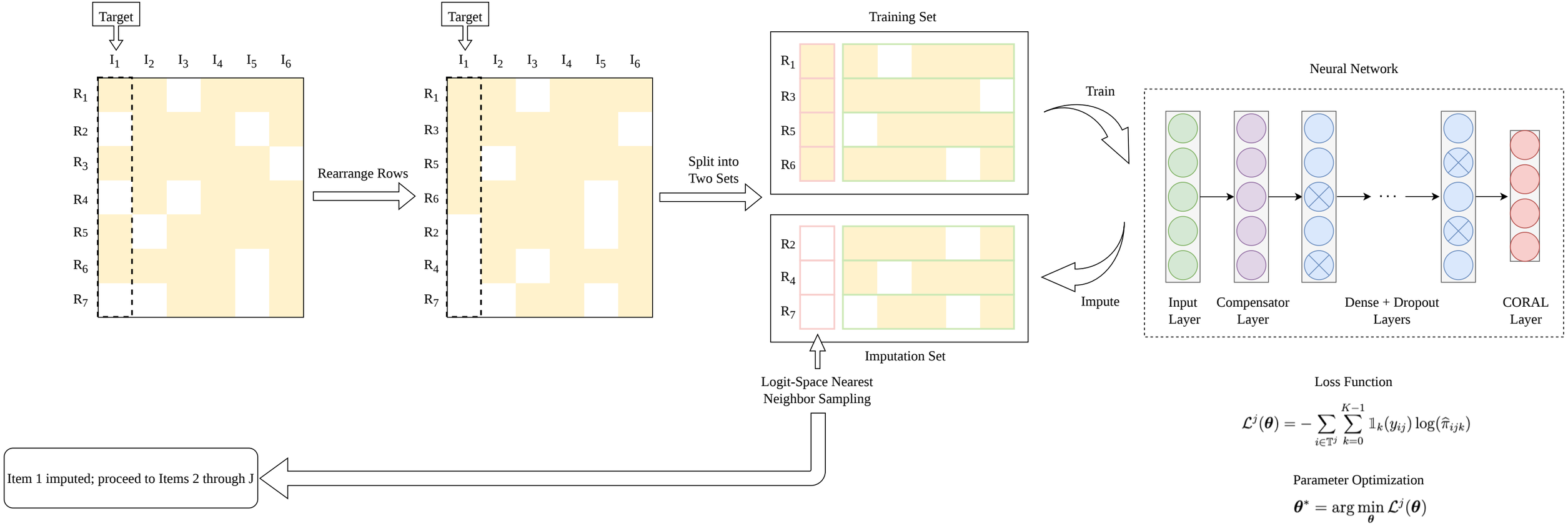

Target
I1 I2 I3 I4 I5 I6
R1 R2 R3 R4 R5 R6 R7
Rearrange Rows
Target
I1 I2 I3 I4 I5 I6
R1 R3 R5 R6 R2 R4 R7
Split into Two Sets
Training Set
R1 R3 R5 R6
R2 R4 R7
Imputation Set
Logit-Space Nearest Neighbor Sampling
Item 1 imputed; proceed to Items 2 through J
Train
Impute
Neural Network
Input Layer
Compensator Layer
Dense + Dropout Layers
CORAL Layer
Loss Function
$\mathcal{L}^j(\boldsymbol{\theta}) = -\sum_{i \in \mathbb{T}^j} \sum_{k=0}^{K-1} \mathbb{1}_k(y_{ij}) \log(\hat{\pi}_{ijk})$
Parameter Optimization
$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} \mathcal{L}^j(\boldsymbol{\theta})$